

Tutorial 3: IS-TR Model

Panthéon-Sorbonne Master in Economics

6th October 2025

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Outline

Core questions

OECD data

Exercises

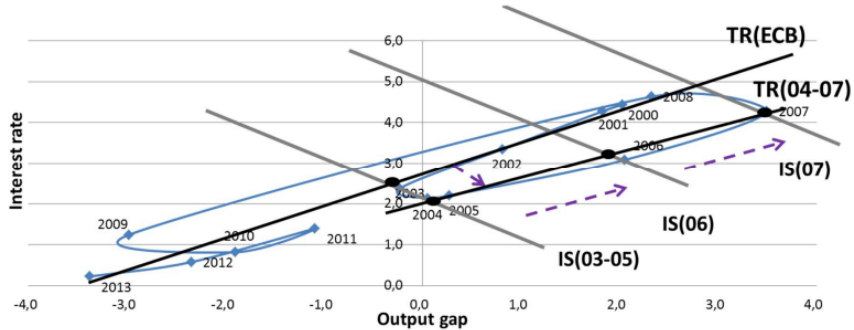
Core questions

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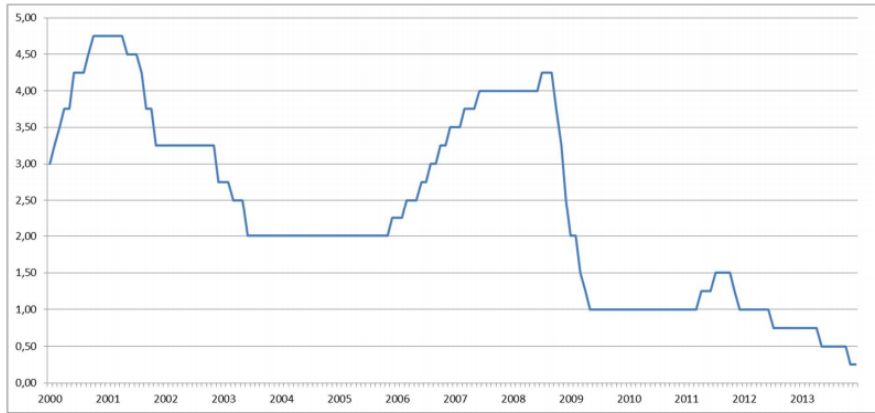
- ▶ What is the crucial Keynesian assumption of the IS-TR model?
- ▶ Why can the desired demand be different from the output?
- ▶ Explain the logic behind Keynesian multiplier.
- ▶ What can cause the shifts of the IS and TR curves?
- ▶ What follows after the shock causing the shift of IS? TR?
(Propagation of the shocks)

OECD data

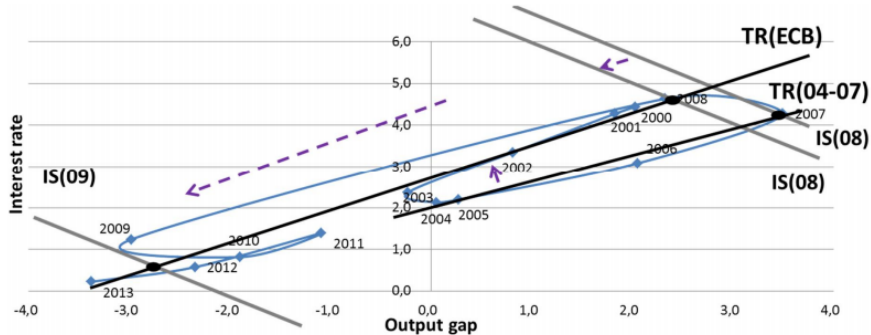
IS-TR model and BC in 2003–2007



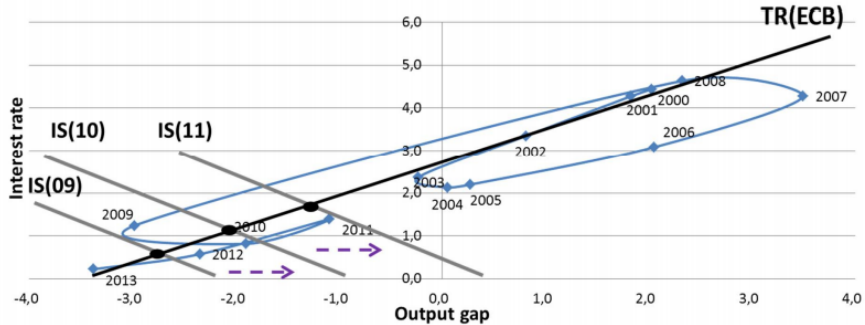
The main interest rate of the ECB



IS-TR model and BC in 2007–2009



IS-TR model and BC in 2009–2011



Exercises

Exercise 1

Desired demand is represented by the following function:

$$DD = 5000 + 0.75(Y - T) + \bar{G} - 100i - 1000\sigma$$

Domestic and foreign price levels have been assumed constant and equal to 1. Let $i = 5\%$ throughout. Initially $\bar{G} = T = 3000$, σ is the nominal exchange rate and is assumed fixed at 1.

1. Compute the effect on GDP of an increase in T from 3000 to 3500. Show your result graphically. What is the value of the tax multiplier?
2. Compute the effect on GDP of an increase in G from 3000 to 3500. Show your results graphically. What is the value of government spending multiplier?
3. Compute the net effect on GDP given both changes combined.

Exercise 2

In reality, tax revenues are not exogenous, but move positively with income/GDP. Suppose that $T = \bar{T} + tY$, where t is an exogenous tax rate. Use this tax revenue function to derive the government spending multiplier in the following simple model:

$$C = c_0 + c_1(Y - T)$$

$$I = \bar{I}$$

$$G = \bar{G}$$

$$PCA = PCA_0 - mY$$

Derive the IS curve for this economy with $I = \bar{I} - 20i$. Plot it in (Y, i) space.

Exercise 3

Suppose that the Taylor rule of an economy is $i = 5 + 0.5(Y - \bar{Y})$, where i is measured in percentage points.

1. Plot the TR curve in (Y, i) space.
2. What is the effect of an increase in $\bar{i}$ from 5 to 6 on the TR curve?
3. Solve for an analytical expression for output (Y) and the interest rate (i) in an equilibrium in an economy with this TR curve and the IS curve from the previous exercise.